

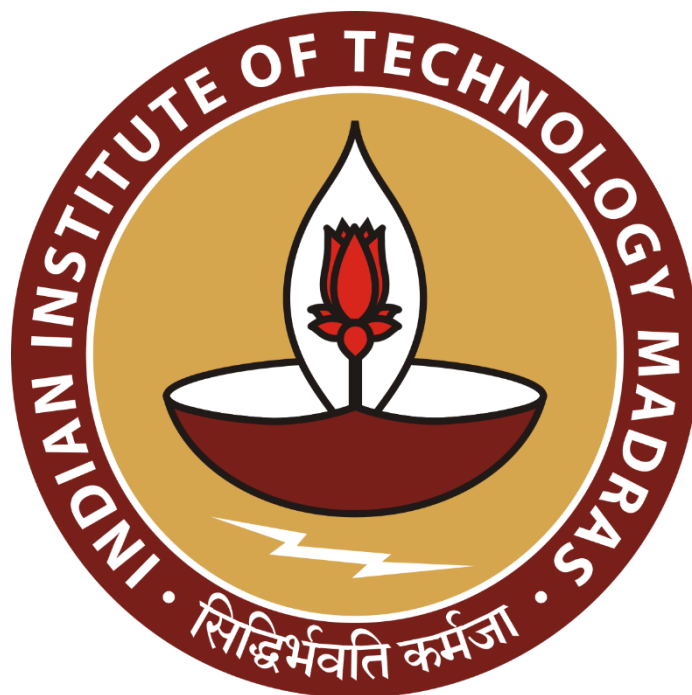
Studying and Enhancing Operational Efficiency at Mahak Spices Company

A Mid-term report for the BDM capstone Project

Submitted by

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1 Executive Summary

Mahak Spices, a regional B2B spice trading shop located at Subhash Nagar, Shahjahanpur, Uttar Pradesh plays a vital role in supplying huge number of spices to local retailers and small businesses. Despite an impactful presence in the wholesale market, the business is currently facing operational challenges that hinders its growth. Main challenges include overdependence on a limited variety of spices, product spoilage during summer months and inconsistent transportation costs.

To address these issues, this report focuses on using data analytics to improve decision-making across sales inventory and logistics. The initial phase involves the collection of data from the business. This data is digitized, cleaned and formatted in spreadsheets for effective data analysis.

Prior analysis indicates uneven sales distribution, with certain spices moving slower comparative to others leading to inefficient use of inventory space. Furthermore, the spice spoilage and transportation cost fluctuations were identified as recurring problems.

The ultimate goal is to build a more adaptive and resilient supply network for Mahak Spices enabling the business to enhance the profitability, minimize spoilage and reduce the fluctuation of transportation costs.

2 Proof of Originality of Data

Shop details –

- Name – Mahak Spices
- Owner's Name – Mahak Kamal Saxena & Akshat Jauhari
- Address - Subhash Nagar, Shahjahanpur, Uttar Pradesh.
- Phone Number – 9044605685
- Type of Business - B2B (Business to Business)
- Number of workers – 3
- Opening time & Closing time – 9:00 AM to 7:00 PM
- Video Link – [Video Link](#)
- Pictures of the shop - [Pictures](#)



Fig 1 – Image of the shop



Fig 2 – Image of the spices

Authorization Letter – *Authorization Letter pdf*

Date: 24/06/2025

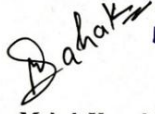
TO WHOMSOEVER IT MAY CONCERN

I hereby grant permission to Ms. Shreya Saxena, a student of the B.S. Degree Program in Data Science & Applications at IIT Madras (Roll No: 22f3001013), to use select business data from **Mahak Spice Company** for her academic capstone project under the course “Business Data Management”.

The data is provided solely for academic use and must be handled with care, confidentiality, and integrity. I confirm that the data shared is authentic to the best of my knowledge and provided in support of her academic work.

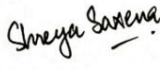
We are pleased to support Ms. Shreya Saxena in her academic journey.

Sincerely,



For Mahak Spice Company

Proprietor
Mahak Kamal Saxena
Owner of the Shop
Mahak Spice Company, Subash Nagar
Shahjahanpur, 242001 Uttar Pradesh



Shreya Saxena
(Roll no – 22f3001013)

Fig 3 – Authorization Letter signed by the proprietor of the shop

3 Meta Data

The data was collected from the co-owner of Mahak Spices, Mr. Akshat Jauhari. The data captures daily records of purchase and sales costs for various spice products over a period of six months (Nov – April).

- Total Rows: 142
- Total Columns: 17
- Purpose:

The data contains the daily raw transactional data for various spices. It includes

information on purchase, sales, quantity sold, cost per kg for each item, and transport charges.

- Excel Sheet link - [Excel Sheet](#)
- Column Headers & Description

Column Name	Description	Data Type
Date	Transaction date	date
Month	Month name	text
Purchase_of_the_day	Total purchase cost for that day	currency
Sales_of_the_day	Total sales amount collected on that day	currency
Turmeric_sold(kg)	Quantity of turmeric sold (in kg)	number
Turmeric_cost_per_kg	Cost per kg of turmeric	currency
RedChilli_sold(kg)	Quantity of red chilli sold (in kg)	number
RedChilli_cost_per_kg	Cost per kg of red chilli	currency
Dhaniya_sold(kg)	Quantity of dhaniya sold (in kg)	number
Dhaniya_cost_per_kg	Cost per kg of dhaniya	currency
CarmonSeeds_sold	Quantity of carom seeds (ajwain) sold (in kg)	number
CarmonSeeds_cost_per_kg	Cost per kg of carom seeds	currency
Cumin_sold	Quantity of cumin (jeera) sold (in kg)	number
Cumin_cost_per_kg	Cost per kg of cumin	currency
Fennel_sold	Quantity of fennel (sauf) sold (in kg)	number
Fennel_cost_per_kg	Cost per kg of fennel	currency
Transport	Transport cost incurred on that day	currency

4 Descriptive Statistics

Using pandas library in python and excel functions that can be used to summarize and describe the main features of a dataset. It gives a quick overview of the data and help us understand its basic characteristics without looking at every single value.

Key Statistics:

Metrics	<i>Turmeric</i>	<i>Red Chilli</i>	<i>Dhaniya</i>	<i>Carmon Seeds (Ajwain)</i>	<i>Cumin (Jeera)</i>	<i>Fennel (Sauf)</i>
<i>Mean</i>	38.92	27.77	24.9	1.56	3.06	1.02
<i>Standard Error</i>	3.67	4.14	1.75	0.5	1.22	0.29
<i>Median</i>	45	0	29	0	0	0
<i>Mode</i>	0	0	0	0	0	0
<i>Standard Deviation</i>	43.69	49.38	20.85	6	14.57	3.49
<i>Sample Variance</i>	1908.65	2438.72	434.59	35.98	212.36	12.19
<i>Kurtosis</i>	7.55	5.39	2.27	36.92	38.01	39.6
<i>Skewness</i>	2.12	2.33	1.15	5.69	5.93	5.71
<i>Range</i>	256	233	95	50	120	30
<i>Minimum</i>	0	0	0	0	0	0
<i>Maximum</i>	256	233	95	50	120	30
<i>Sum</i>	5527	3943	3536	222	435	145
<i>Count</i>	142	142	142	142	142	142

Fig 4 - Descriptive Statistics Summary of Spices sales data

1. Turmeric

The average daily sales for Turmeric are 38.92 kgs, also they have a maximum sale of 256.00 kgs, indicating highly variable demand. The standard deviation of 43.69 kgs shows fluctuations that are substantial. Often it is bulk purchases or special events that drive them. Most days see low to moderate sales volumes while the positive skewness of 2.12 points to occasional days with exceptionally high sales.

2. Red Chilli

The average daily sales of Red Chilli are 27.77 kgs, and sales range from 0.00 to 233.00 kgs which highlights important variability. Demand spikes frequently since a standard deviation of 49.38 kgs suggests it. Spikes perhaps connect to festival or seasonal needs. For most days, there are modest sales; however, the 2.33 skewness indicates that on occasional days, there are very high sales.

3. Dhaniya

The average daily sales for Dhaniya are 24.9 kgs, also they have a maximum sales of 95.00 kgs, reflects moderate variability. Most days are stable, though some days

have larger sales, as the standard deviation suggests with 20.85 kgs. A skewness of 1.15 does suggest something now. It suggests the notion that occasional high sales days have a tendency to occur.

4. Carmon Seeds (Ajwain)

The average daily sales for Carmon Seeds (Ajwain) are about 1.56 kgs, and they range in value from 0.00 to 50.00 kgs, showing some sporadic demand. High variability in relation to the mean is pointed to by the 6.00 kgs standard deviation, as spikes within sales are meaningful yet rare. A high skewness of 5.69 exists because it indicates few or no sales occur most days, yet large quantities get purchased a few days.

5. Cumin (Jeera)

The average daily sales of Cumin (Jeera) are 3.06 kgs, and sales go from 0.00 to 120.00 kgs, showing big change. A value that is a standard deviation of 14.57 kgs shows that, for most days, sales are low. However, at times deals do turn quite large. The 5.93 skewness shows a strong trend toward uncommon days that sell a lot.

6. Fennel (Sauf)

The average daily sales for Fennel (Sauf) are 1.02 kgs, with a maximum sale of 30.00 kgs, indicating low but variable demand. The standard deviation of 3.49 kgs highlights that most days have minimal sales, with occasional spikes. The skewness of 5.71 suggests that high sales days are rare but impactful.

5 Detailed Explanation of Analysis Process

I. Data Cleaning and formatting

The data that was collected was first converted into excel format for ease of access and manipulation. Once in Excel, the data is cleaned and formatted to ensure consistency and usability. Cells with missing or incomplete data and redundancy were carefully reviewed and resolved. Standardizing all the columns with the specific

datatypes. After data cleaning the data was formatted for further analysis, for example (naming all the columns with standard names “Turmeric_sold”).

II. Descriptive Statistics Analysis

Some Calculations such as the revenue of each spice is calculated using the formula “Revenue = Spice_sold x Cost_per_kg”. Same way the profit per day was calculated by the formula “Profit per day = Sales_per_day – Purchase_per_day” . Key statistical measures—such as mean, median, mode, range, standard deviation, and percentiles—were calculated for each spice.

Descriptive statistics played a major role within clearly and concisely summarizing and interpreting each spice's sales behavior. These stats described data usefully when combined as one base. Business decisions were driven by data as well as involved spotting irregularities plus identifying trends.

III. Profitability Analysis

Profit for each spice was calculated on a daily basis by subtracting the total cost (quantity sold × cost per kilogram) from the total sales revenue for that day. These daily profit values were aggregated to obtain monthly profit figures for each spice. For each product, organizing and summing up sales and cost data was the analysis. This dual-level analysis—both day-wise and month-wise—allowed for detailed tracking of performance over time. For identification of which spices were consistently the most and least profitable, the monthly profit results were then compared across different spices and time periods.

IV. Tools Used:

- Python – Data Cleaning, Data Processing and Analysis
- Excel – Manual Calculations and Formula, Data Presentation

6 Results and Findings

The following result is obtained on performing the above analysis.

1. Spice-wise share contribution in the total revenue

- The highest contributor is Turmeric with 42 % of the total revenue followed by Red Chilli with 33% and then Dhaniya.
- The chart visually stresses the heavy revenue reliance on Red Chilli and Turmeric along with the other spices that together contribute less than one-fifth of the total.

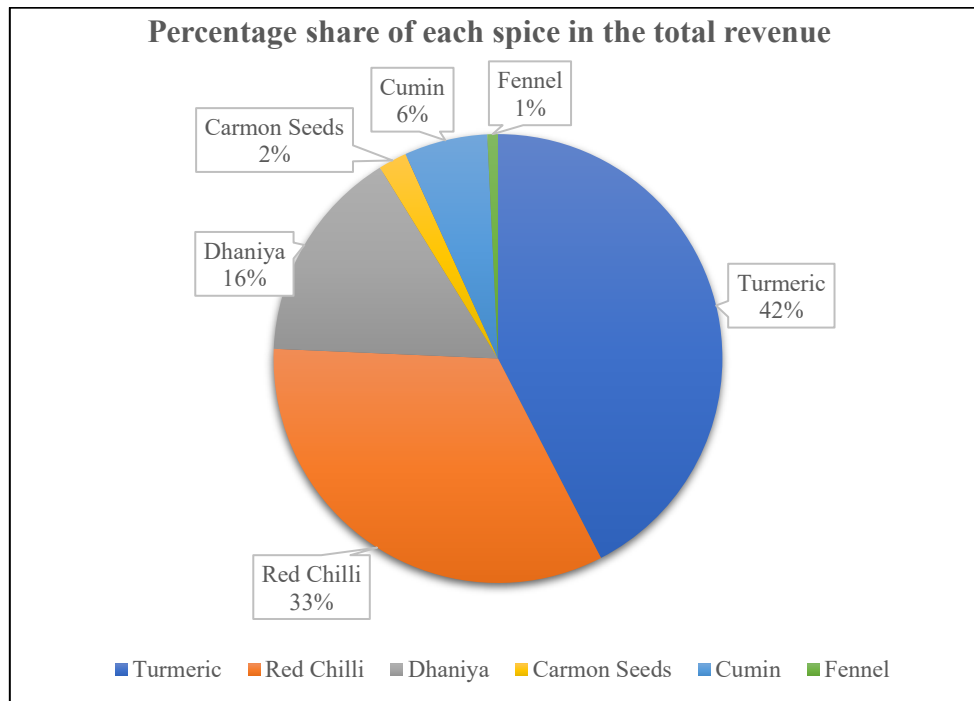


Fig 5 – Pie Chart of the contribution of each spice in total revenue

2. Monthly Revenue Comparison by spice

- Turmeric and Red Chilli are the Top Earners: Compared to other spices Turmeric and Red Chilli generate the highest revenues with notable peaks in February and March. The revenue for turmeric peaks in March and in April which is a trend that is prominent. November, December, also February are when Red Chilli leads instead.
- For revenue generation, revenue generation has a clear seasonal trend now. Red Chilli sees its highest revenue within February, whereas Turmeric's revenue rises so sharply during March as well as April. Dhaniya, Cumin (Jeera), and Fennel (Sauf) revenues remain stable though lower than others throughout the period.
- Across all months, Carmon Seeds (Ajwain) consistently generate the least revenue among the spices. For Carmon Seeds (Ajwain), this suggests lower sales volume or lower pricing.

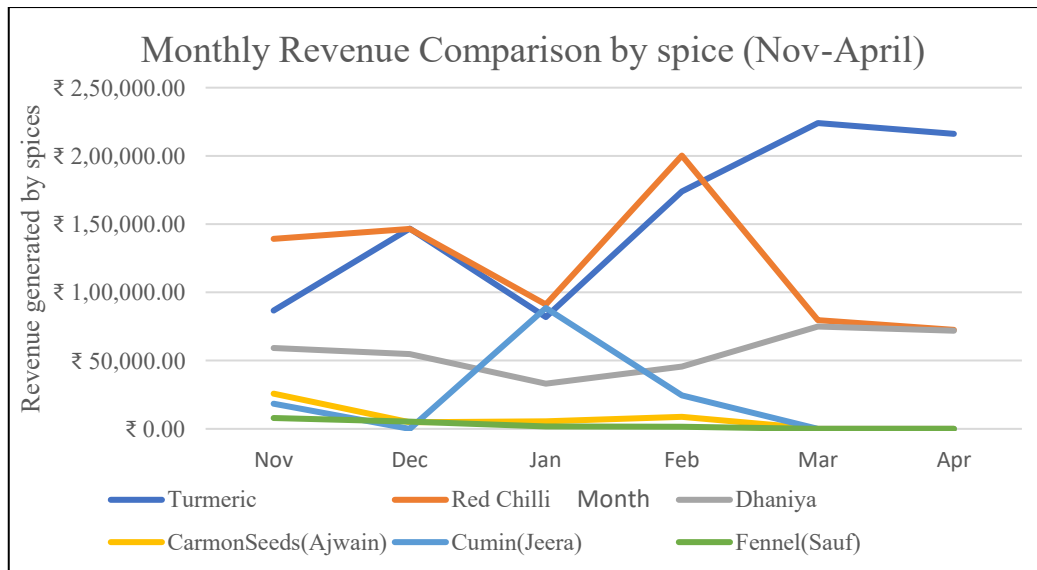


Fig 6 – Line Chart for the monthly revenue trend of each spice over the months

3. Profit trends over months

- The profit saw a sharp decline from November into December because it indicated possible increased costs, reduced sales during that period.
- Profits recovered in January after the December dip because they signaled business recovery or better sales performance.
- November recorded the highest profit within the six months. This suggests that it was because of strong sales or of efficient cost management at the period's start.

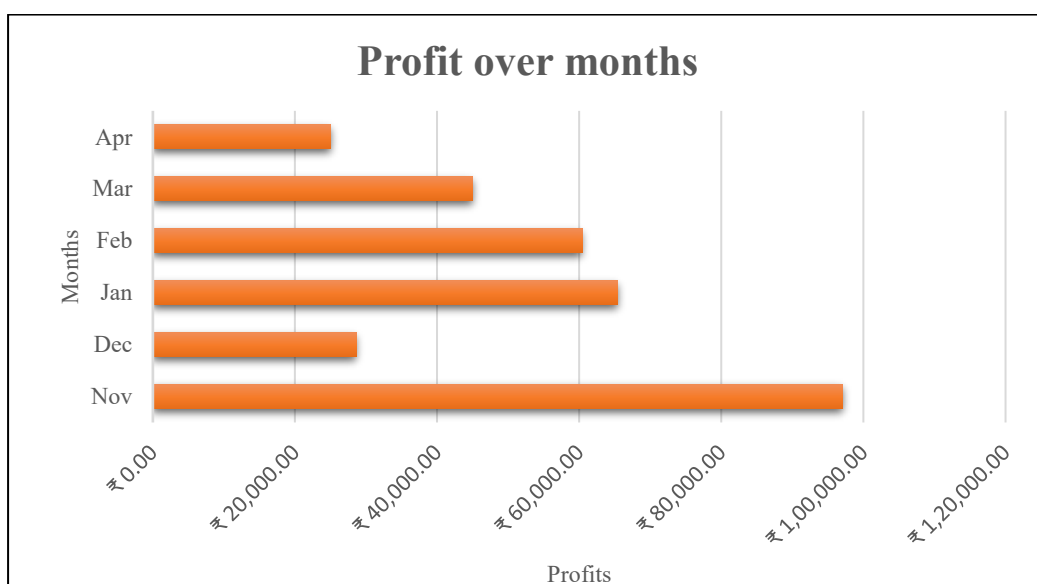


Fig 7 – Bar Chart for the monthly profit